DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CYBER SECURITY)

Seminar Report
on

AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

Submitted in partial fulfillment for the award of the Degree of

Bachelor of Engineering
in
Computer Science and Engineering (Cyber Security)

Submitted by

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Under the Guidance of

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2025-26

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CERTIFICATE

Certified that the seminar report entitled **"AI Assisted Trust Evaluation In Blockchain Based Digital Twin System**" carried out by **Parvati Waladunki** bearing a **USN:1DS22CY032** a bonafide student of **DAYANANDA SAGAR COLLEGE OF ENGINEERING**, an autonomous institution affiliated to VTU, Belagavi in partial fulfillment for the award of Degree of **Bachelor of Engineering in Computer Science & Engineering (Cyber Security**) during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Seminar report has been approved as it satisfies the academic requirements with respect to the work prescribed for the said Degree.

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DECLARATION

I, **Parvati Waladunki (1DS22CY032),** hereby declare that the seminar work entitled " AI Assisted Trust Evaluation In Blockchain Based Digital Twin System " has been independently done by me under the guidance of **'Dr. Deepthi V S', Assistant Professor, department of CSE(Cyber Security)** and submitted in partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering in Computer Science & Engineering (Cyber Security)** at **Dayananda Sagar College of Engineering**, an autonomous institution affiliated to VTU, Belagavi during the academic year 2025-2026.

I further declare that I have not submitted this report either in part or in full to any other university for the award of any degree.

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ACKNOWLEDGEMENT

The satisfaction and euphoria accompanying the successful completion of any task would be incomplete without the mention of people who made it possible under constant guidance and encouragement. I sincerely thank the **Management of Dayananda Sagar College of Engineering, Bengaluru.**

I express our sincere regards and thanks to **Dr. B G Prasad, Principal, Dayananda Sagar College of Engineering, Bengaluru.** His constant encouragement guidance and valuable support have been an immense help in realizing the seminar.

I express my sincere regards and thanks to **Dr. Mohammed Tajuddin, Professor & Head, Department of Computer Science and Engineering (Cyber Security), Dayananda Sagar College of Engineering, Bengaluru.** His incessant encouragement guidance and valuable technical support have been an immense help in realizing this project. His guidance gave us the environment to enhance our knowledge, and skills and to reach the pinnacle with sheer determination, dedication, and hard work.

I would like to express profound gratitude to my guide **Dr. Deepthi V S, Assistant Professor, Department of Computer Science and Engineering (Cyber Security), Dayananda Sagar College of Engineering, Bengaluru** who has encouraged me throughout the process. **Her** moral support enabled me to complete the work successfully.

I thank all teaching and non-teaching staff of the Department of Computer Science and Engineering (Cyber Security), for their kind and constant support throughout the academic journey.

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ABSTRACT

The rapid growth of cyber-physical systems and Internet of Things (IoT) environments has led to the increasing adoption of digital twin technology for real-time monitoring, analysis, and decision-making. However, ensuring data integrity, reliability, and trustworthiness remains a critical challenge due to the presence of malicious data sources, system vulnerabilities, and centralized control limitations. This seminar focuses on an AI-assisted trust evaluation framework integrated within a blockchain-based digital twin system to address these challenges.

The primary objective of this work is to develop a secure and intelligent system that evaluates the trustworthiness of data generated by physical entities before incorporating it into the digital twin. The proposed approach combines artificial intelligence techniques, such as anomaly detection and machine learning-based trust scoring, with blockchain technology to provide a decentralized and tamper-resistant data storage mechanism. The system collects real-time data from sensors, processes it through an AI-based trust evaluation module, and securely records validated data on a blockchain network.

The results of this approach demonstrate improved data reliability, enhanced system security, and efficient detection of anomalous or malicious inputs. By leveraging blockchain's immutability and AI's predictive capabilities, the system ensures that only trusted data contributes to the digital twin, thereby improving decision-making accuracy.

This framework has significant applications in smart cities, industrial automation, healthcare monitoring, and intelligent transportation systems, where secure and trustworthy data is essential. Future scope includes the integration of advanced AI models such as deep learning and federated learning, scalability improvements for large-scale deployments, and optimization of blockchain performance for real-time applications.

Keywords: *Digital twin, Blockchain, Trust Evaluation, Artificial Intelligence, Data Security, Decentralized Systems*

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LIST OF ABBREVIATIONS

Abbreviation	Full Description
AI	Artificial Intelligence
DT	Digital Twin
IoT	Internet of Things
IIoT	Industrial Internet of Things
ML	Machine Learning
DL	Deep Learning
FL	Federated Learning
DLT	Distributive Ledger Technology
PoA	Proof of Authority
PoW	Proof of Work
PoS	Proof of Stake
SC	Smart Contracts
API	Application Programming Interface
LSTM	Long Short Term Memory
IDS	Intrusion Detection System
IAM	Identity and Access Management
CPU	Central Processing Unit
GPU	Graphical Processing Unit
SLA	Service Level Agreement

CHAPTER 1

INTRODUCTION

1.1 Overview of AI-Assisted Trust Evaluation in Blockchain-Based Digital Twin Systems

The convergence of Digital Twin technology, Blockchain, and Artificial Intelligence (AI) represents a transformative advancement in modern computing systems, particularly within the context of Industry 4.0 and smart environments. A Digital Twin is a virtual representation of a physical entity or system that continuously receives real-time data from sensors and connected devices. This virtual model enables simulation, monitoring, predictive analysis, and intelligent decision-making. However, as these systems rely heavily on real-time data streams, ensuring the trustworthiness and integrity of data becomes a critical concern.

Blockchain technology addresses these concerns by providing a decentralized and immutable ledger that records transactions in a secure and transparent manner. It eliminates the need for centralized authorities and ensures that data cannot be altered once recorded. Despite this, blockchain alone does not inherently evaluate the quality or reliability of incoming data. This is where Artificial Intelligence plays a vital role. AI-based trust evaluation mechanisms analyze data patterns, detect anomalies, and assign trust scores, enabling systems to distinguish between reliable and potentially malicious data inputs.

The integration of these three technologies creates a robust framework where AI evaluates trust, blockchain secures data, and digital twins utilize trusted data for accurate modeling and predictions. This integrated system is particularly relevant in environments such as smart cities, industrial automation, healthcare systems, and intelligent transportation, where decision-making depends on accurate and trustworthy data.

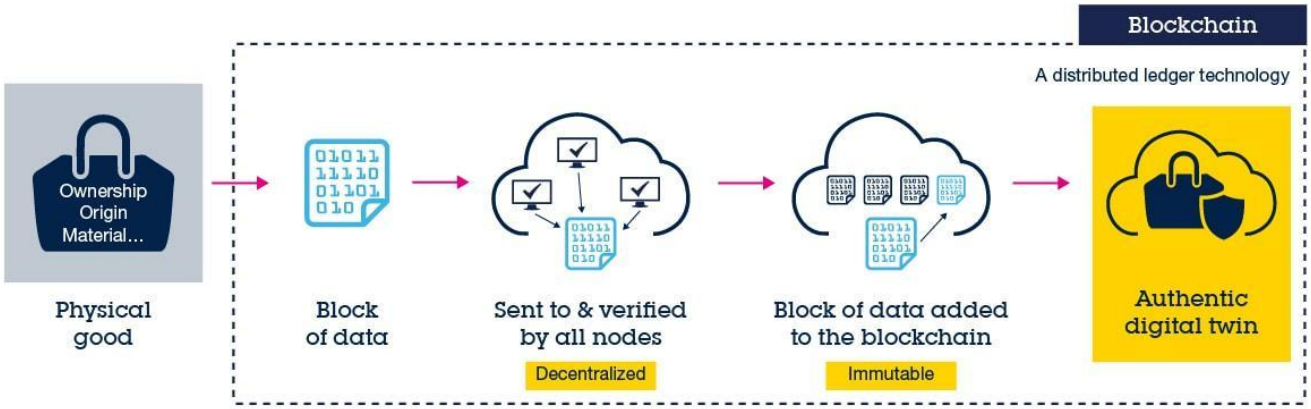

Fig 1.1.1 Blockchain Based Digital Twin System

1.2 Evolution of the Technology

The concept of digital twins originated in the early 2000s, initially employed in manufacturing and aerospace industries to mirror physical assets and processes. Over the years, as IoT devices proliferated, digital twins expanded beyond individual machines to complex systems—such as smart cities, healthcare infrastructure, and industrial operations. Alongside this evolution, ensuring the security and trustworthiness of the data feeding these digital replicas became paramount. Traditional centralized systems posed risks: single points of failure, susceptibility to tampering, and limited transparency.

The emergence of blockchain originally conceptualized for cryptocurrencies provided a decentralized, tamper-proof solution for recording transactions and data. More recently, the integration of AI has revolutionized trust evaluation, allowing automated, adaptive assessment of data reliability, anomaly detection, and risk mitigation. Thus, the convergence of digital twins, blockchain, and AI represents the latest stage in ensuring that complex, interconnected systems are both intelligent and secure, evolving from basic replication to trusted, real-time decision-making ecosystems.

The concept of Digital Twins originated from the field of product lifecycle management and was initially used in aerospace and manufacturing industries to simulate and monitor physical systems.

Over time, with the advancement of the Internet of Things (IoT), Digital Twins evolved into dynamic systems capable of real-time synchronization with physical entities through continuous data streams. This evolution enabled more sophisticated applications such as predictive maintenance, performance optimization, and system simulation.

Parallel to this, blockchain technology emerged with the introduction of cryptocurrencies, most notably Bitcoin. Initially designed as a decentralized financial system, blockchain has since evolved into a versatile technology used across various domains for secure data management, smart contracts, and decentralized applications. Its ability to ensure data immutability and transparency made it an ideal candidate for securing data generated by Digital Twins.

Artificial Intelligence has also undergone significant transformation, evolving from rule-based systems to advanced machine learning and deep learning models capable of analyzing large volumes of data with high accuracy. The integration of AI into trust evaluation mechanisms has enabled systems to autonomously detect anomalies, predict potential threats, and assess the reliability of data sources.

The convergence of these technologies marks a significant milestone, where Digital Twins provide real-time insights, blockchain ensures data security, and AI enhances trust and intelligence. This integrated approach represents the next generation of intelligent and secure cyber-physical systems.

1.3 Importance of AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

In today's increasingly connected world, digital twins are pivotal in optimizing real-world systems from smart grids and autonomous vehicles to predictive healthcare and supply chains. However, these systems are only as effective as the data that drives them. Data breaches, malicious inputs, or even unintentional inaccuracies can lead to significant consequences whether it's incorrect predictions, operational failures, or compromised safety. Ensuring trust in these environments is critical. Blockchain's immutable ledger provides the backbone for secure data sharing, preventing manipulation or unauthorized changes.

Meanwhile, AI-based trust evaluation ensures that only high-integrity data is fed into digital twin models, filtering out anomalies or suspicious patterns in real time. This combination is not merely a technical improvement it's foundational for industries where safety, accuracy, and security are paramount. Whether it's in industrial automation, where predictive maintenance prevents costly downtime, or in healthcare, where patient monitoring relies on reliable data, ensuring trust enhances efficiency, reduces risk, and builds confidence in autonomous systems. As digital twins become central to future infrastructure, the ability to secure and trust them determines their viability in mission-critical environments.

In modern interconnected systems, the reliability of data directly influences the effectiveness of decision-making processes. Digital Twin systems depend on continuous data input from sensors and external sources, making them vulnerable to data manipulation, cyber-attacks, and system failures. Without proper trust evaluation mechanisms, these systems may operate on inaccurate or malicious data, leading to incorrect predictions and potentially catastrophic outcomes.

The integration of AI-assisted trust evaluation addresses this issue by enabling systems to assess the credibility of data in real time. AI models can identify abnormal patterns, detect inconsistencies, and evaluate the behavior of data sources. When combined with blockchain, which ensures that validated data is securely stored and cannot be altered, the system achieves a high level of trust and reliability.

This approach is particularly important in critical domains such as healthcare, where inaccurate data can affect patient outcomes; industrial automation, where system failures can lead to significant financial losses; and smart cities, where large-scale infrastructure depends on accurate data for efficient operation. Therefore, the proposed system plays a crucial role in enhancing security, reliability, and operational efficiency.

1.4 Scope of AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

The scope of AI-assisted trust evaluation in blockchain-based digital twin systems is extensive and spans multiple domains. This approach can be applied in smart cities for traffic management and

infrastructure monitoring, in healthcare for patient monitoring and predictive diagnostics, in industrial IoT for predictive maintenance, and in supply chain management for ensuring transparency and traceability.

The integration of AI allows the system to handle large volumes of data and adapt to changing conditions, while blockchain provides a secure and decentralized platform for data storage and sharing. The scalability and flexibility of this approach make it suitable for a wide range of applications where data integrity and trust are critical.

The scope of this study encompasses the design and analysis of a framework that integrates Digital Twin technology with blockchain and AI-based trust evaluation mechanisms. It focuses on understanding how these technologies can work together to address challenges related to data security, trust, and system reliability in distributed environments.

The study covers various application domains, including industrial IoT, smart cities, healthcare systems, and supply chain management. In industrial settings, the system can be used for predictive maintenance and process optimization. In healthcare, it can enable real-time patient monitoring with secure and trustworthy data handling. In smart cities, it can support intelligent infrastructure management, traffic control, and energy optimization.

Furthermore, the study explores the role of AI in evaluating trust and detecting anomalies, as well as the use of blockchain for secure data storage and decentralized trust management. While the focus is on conceptual and architectural understanding, the framework can be extended to practical implementations in real-world systems.

1.5 Objectives of AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

The primary objective is to explore and understand the role of artificial intelligence in evaluating trust within blockchain-based digital twin systems. It aims to analyze how the integration of these technologies can enhance data security, reliability, and system performance. Another objective is to study the methodologies used for trust evaluation, including anomaly detection and trust scoring mechanisms.

The work seeks to identify the benefits and limitations of the proposed approach and to highlight its potential applications in real-world scenarios. Ultimately, the goal is to provide a comprehensive understanding of how AI-assisted trust evaluation can address the challenges faced by modern digital twin systems.

Develop a comprehensive understanding of how Artificial Intelligence can be utilized to evaluate trust within blockchain-based Digital Twin systems. It aims to analyze the limitations of traditional systems and propose an integrated approach that enhances data reliability and system security.

Another objective is to explore the role of blockchain in ensuring data integrity and transparency, while also examining how AI models can improve trust evaluation through anomaly detection and predictive analysis. The study also seeks to identify suitable methodologies and frameworks that can be applied in real-world scenarios.

Additionally, the work aims to highlight the potential benefits and applications of the proposed system across various domains. It also focuses on understanding the challenges involved in implementing such systems and identifying possible solutions to overcome them. Ultimately, the objective is to contribute to the development of secure, intelligent, and trustworthy digital ecosystems.

1.6 Challenges of AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

Despite its advantages, the integration of AI, blockchain, and Digital Twin technologies presents several challenges. One of the major challenges is scalability, as blockchain networks can experience performance issues when handling large volumes of data generated by Digital Twin systems. The latency associated with blockchain transactions may also affect real-time operations.

Another challenge is the quality and reliability of data used for training AI models. If the training data is biased or inaccurate, the trust evaluation results may be compromised. Additionally, AI models require significant computational resources, which may not be feasible in resource-constrained environments.

Security is another critical concern, as IoT devices used in Digital Twin systems are often vulnerable to cyber-attacks. Ensuring secure communication between devices, AI modules, and blockchain networks is essential. Furthermore, integrating these technologies into a unified system requires careful design and coordination, which can be complex and resource-intensive.

1.7 System Requirements for AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

To implement an effective AI-assisted trust evaluation system, several requirements must be fulfilled. The system must have robust data collection mechanisms, including reliable sensors and communication networks. It should support advanced AI algorithms capable of analyzing large datasets and detecting anomalies in real time. A secure and scalable blockchain infrastructure is required to store and manage data efficiently. Furthermore, the system must ensure interoperability between different components and support seamless integration with existing technologies. Adequate computational resources and efficient data management strategies are also essential to handle the complexity of the system.

The implementation of an AI-assisted trust evaluation system in a blockchain-based Digital Twin environment requires a combination of hardware and software components. On the hardware side, IoT sensors and devices are needed to collect real-time data from physical systems. These devices must be capable of secure communication and reliable data transmission.

On the software side, the system requires a Digital Twin platform for modeling and simulation, AI algorithms for trust evaluation, and a blockchain network for secure data storage and management. The integration of these components requires middleware and communication protocols that ensure seamless data flow and interoperability.

Additionally, the system must support scalability, low latency, and high reliability to handle large volumes of data and real-time processing requirements. Security mechanisms such as encryption, authentication, and access control are also essential to protect the system from unauthorized access and cyber threats.

CHAPTER 2

LITERATURE SURVEY

2.1 Zhang Y., Liu X., Wang H.: Blockchain-Based Trust Mechanism for Digital Twin in Industrial IoT

Context and Problem

Industrial Internet of Things (IIoT) environments rely on a large number of interconnected devices and sensors that continuously generate data for digital twin systems. These digital twins use real-time data to monitor operations, predict failures, and optimize performance. However, the reliability of such systems is highly dependent on the trustworthiness of incoming data. In traditional architectures, data is often managed through centralized servers, which are vulnerable to cyber-attacks, data tampering, and single points of failure. Additionally, there is no effective mechanism to verify whether the data generated by sensors is genuine or manipulated. This creates a significant challenge in maintaining data integrity and trust within digital twin systems, especially in critical industrial applications where incorrect data can lead to operational failures and financial losses.

Methodology and Contribution

To address these challenges, the authors propose a blockchain-based trust mechanism specifically designed for digital twin systems in IIoT environments. The methodology involves integrating a decentralized blockchain network with digital twin architecture to ensure secure and transparent data management. A Proof of Authority (PoA) consensus mechanism is used to reduce computational overhead and improve efficiency, making it suitable for industrial applications. The system evaluates trust by analyzing multiple parameters such as data consistency, historical reliability of devices, and behavioral patterns of nodes. Once the data is validated, it is stored on the blockchain, ensuring immutability and traceability. Smart contracts are utilized to automate the trust evaluation process and enforce predefined rules for data validation. The key contribution of this work lies in developing a lightweight and efficient trust framework that combines

blockchain with digital twin systems, enabling secure, decentralized, and reliable data exchange in IIoT environments.

Key Findings

- The proposed system significantly enhances data integrity by preventing unauthorized modifications.
- Use of PoA consensus reduces computational cost and improves system efficiency, decentralized architecture eliminates single points of failure.
- Trust evaluation based on multiple parameters improves accuracy of data validation and smart contracts enable automated and consistent trust management.

Conclusion

The study concludes that integrating blockchain with digital twin systems provides a robust solution for addressing trust and security challenges in Industrial IoT environments. By ensuring data immutability, transparency, and decentralized validation, the proposed framework improves the reliability of digital twin operations. The use of efficient consensus mechanisms and automated trust evaluation further enhances system performance, making it suitable for real-world industrial applications.

2.2 Chen J., Li Q., Zhao K.TFL-DT.: A Trust Evaluation Scheme for Federated Learning in Digital Twin Systems

Context and Problem

The rapid integration of federated learning with digital twin systems has enabled collaborative model training without sharing raw data, thereby preserving privacy. However, this decentralized learning paradigm introduces significant challenges related to trust, as multiple participants contribute local model updates that may be unreliable, noisy, or even malicious. In digital twin environments, where decisions are made based on continuously updated models, the presence of untrustworthy participants can degrade model performance and compromise system reliability. The core problem addressed in this work is the absence of an effective trust evaluation mechanism

to identify and filter out low-quality or malicious contributions in federated learning-based digital twin systems.

Methodology and Contribution

To address this issue, the authors propose a Trust-aware Federated Learning framework for Digital Twins (TFL-DT) that incorporates a dynamic trust evaluation mechanism. The methodology involves assigning a trust score to each participant based on multiple factors, including the quality of their model updates, historical contribution consistency, and behavioral patterns over time. Machine learning techniques are used to analyze deviations in model updates and detect anomalies that may indicate malicious intent or poor data quality. The system continuously updates trust scores and adjusts the weight of each participant's contribution during the aggregation process. Participants with low trust scores are either penalized or excluded from the training process. The key contribution of this work lies in integrating adaptive trust evaluation directly into the federated learning workflow within a digital twin environment, ensuring that only reliable data influences model updates.

Key Findings

- The proposed trust evaluation mechanism significantly improves the accuracy and robustness of federated learning models.
- Malicious or low-quality participants are effectively identified and filtered out, dynamic trust scoring enhances adaptability to changing participant behaviour.
- The system maintains data privacy while improving trust and reliability, reduced impact of poisoning attacks and unreliable data contributions.

Conclusion

The paper concludes that incorporating AI-driven trust evaluation into federated learning frameworks is essential for maintaining the integrity and performance of digital twin systems. By dynamically assessing participant reliability and controlling their influence on model training, the TFL-DT approach ensures secure, accurate, and trustworthy collaborative learning in decentralized environments.

2.3 Kumar A., Singh R., Patel S.: Blockchain-Enabled Digital Twin Framework for Smart Cities

Context and Problem

Traditional centralized architectures used in smart cities suffer from several limitations, including vulnerability to cyber-attacks, lack of transparency, data inconsistency, and risks of data tampering. The key problem addressed in this work is the absence of a secure and trustworthy mechanism for managing and validating data within smart city digital twin systems. Without proper trust evaluation, inaccurate or malicious data can propagate through the system, leading to incorrect predictions and inefficient decision-making. Therefore, there is a need for a framework that ensures data integrity, enhances trust, and supports intelligent analysis in a decentralized environment.

Methodology and Contribution

The authors propose a blockchain-enabled digital twin framework integrated with Artificial Intelligence, specifically using Long Short-Term Memory (LSTM) networks for predictive analysis. The methodology involves collecting real-time data from various smart city components such as traffic sensors, environmental monitoring systems, and public infrastructure devices. This data is first processed using AI models to analyze patterns, detect anomalies, and evaluate trustworthiness. Once the data is validated, it is stored on a blockchain network, ensuring immutability, transparency, and decentralized control. Blockchain also enables secure data sharing among different stakeholders without relying on a central authority. Smart contracts are utilized to automate data validation and enforce predefined rules for trust management. The major contribution of this work lies in the integration of AI-based predictive analytics with blockchain-based security mechanisms in a digital twin environment. This combination enhances both the intelligence and security of smart city systems, providing a reliable foundation for real-time decision-making.

Key Findings

- The integration of LSTM models significantly improves prediction accuracy in smart city applications.

- Blockchain ensures data integrity and prevents unauthorized data modification, the system enhances transparency and enables secure data sharing across multiple entities.
- AI-based analysis effectively detects anomalies and unreliable data sources, the combined framework increases overall trust in digital twin-based smart city systems.

Conclusion

The paper concludes that the integration of blockchain and AI within digital twin systems provides a robust solution to address the challenges of data trust and security in smart cities. By combining intelligent data analysis with decentralized and tamper-proof storage, the proposed framework ensures reliable and efficient system operations. The study demonstrates that such an approach can significantly enhance the performance, transparency, and trustworthiness of smart city infrastructures, making it a promising direction for future research and real-world implementations.

2.4 Rahman M., Islam S., Ahmed T.: Blockchain-Based Collaborative Digital Twin Framework for Healthcare

Context and Problem

Healthcare environments are inherently collaborative, involving multiple stakeholders such as hospitals, laboratories, insurance providers, and regulatory bodies. This creates a complex ecosystem where sensitive medical data must be shared across different entities. The major problem identified in this context is the lack of secure, trustworthy, and privacy-preserving data sharing mechanisms. Traditional centralized systems are prone to data breaches, unauthorized access, and single points of failure. Additionally, ensuring data integrity and patient privacy while enabling real-time collaboration remains a significant challenge. There is also a lack of transparency and traceability in how patient data is accessed and used across different entities.

Methodology and Contribution

The framework operates by collecting real-time patient data from IoT-enabled medical devices and feeding it into digital twin models for simulation and analysis. Before this data is shared or stored, it undergoes validation and encryption processes to ensure confidentiality. Blockchain technology is then used as a distributed ledger to store hashes of the data, ensuring immutability,

transparency, and traceability. Smart contracts are implemented to define access control policies, allowing only authorized entities to access specific patient data. These contracts automate trust management by verifying identities and enforcing predefined rules without the need for a central authority. The system also incorporates identity management techniques to ensure that each participant in the network is authenticated and accountable. The key contribution of this work lies in providing a secure, decentralized, and collaborative framework that enables multiple healthcare entities to interact with digital twins while maintaining data privacy, integrity, and trust.

Key Findings

- Blockchain ensures tamper-proof storage of healthcare data, improving data integrity, smart contracts enable automated and secure access control mechanisms.
- The framework supports secure collaboration among multiple healthcare entities and patient data privacy is preserved through encryption and decentralized storage.
- Transparency and traceability of data access are significantly improved and the system reduces dependency on centralized infrastructure, minimizing single points of failure.

Conclusion

The study concludes that integrating blockchain with digital twin technology provides an effective solution for addressing trust, security, and privacy challenges in healthcare systems. The proposed framework enables secure and efficient collaboration among different entities while ensuring that sensitive medical data remains protected and trustworthy. This approach has strong potential for real-world implementation in smart healthcare environments, particularly in scenarios requiring remote monitoring and multi-party data sharing.

2.5 Wang L., Zhou Y., Chen H.: Causio-TwinChain: AI-Driven Trust and Security Model for Digital Twins

Context and Problem

Digital twin systems rely on continuous data streams from sensors and external sources to simulate and optimize real-world processes. However, these systems face significant challenges in ensuring data trustworthiness, as malicious inputs, noisy data, or unexpected variations can compromise

system accuracy. Traditional anomaly detection techniques used in such systems often depend on correlation-based models, which are limited in their ability to distinguish between genuine anomalies and normal fluctuations. This leads to high false positive rates and unreliable trust evaluation. The paper identifies the need for a more intelligent and context-aware trust evaluation mechanism that can accurately interpret system behavior and detect threats.

Methodology and Contribution

To address these challenges, the authors propose a novel framework that combines causal inference models with blockchain technology. Unlike traditional machine learning approaches, causal AI analyzes cause-and-effect relationships between variables, enabling the system to understand the underlying reasons behind data changes rather than just identifying patterns.

The methodology involves constructing a causal model of the digital twin system, where relationships between different parameters are defined. Incoming data is evaluated against this model to detect inconsistencies and anomalies based on causal reasoning. Once the data is verified, it is stored on a blockchain network to ensure immutability, transparency, and secure sharing. The system also incorporates a trust evaluation mechanism that assigns trust scores based on the consistency and reliability of data sources.

The key contribution of this work lies in introducing causal AI into trust evaluation for digital twins, significantly improving the accuracy of anomaly detection and reducing reliance on traditional correlation-based methods.

Key Findings

- Significant reduction in false positive rates compared to traditional anomaly detection methods.
- Improved accuracy in identifying genuine anomalies using causal relationships, enhanced trust evaluation through context-aware analysis.
- Secure and tamper-proof data storage using blockchain integration and increased reliability and robustness of digital twin systems.

Conclusion

The paper concludes that integrating causal AI with blockchain provides a more intelligent and reliable approach to trust evaluation in digital twin systems. By focusing on cause-and-effect relationships, the proposed framework overcomes the limitations of traditional methods and improves anomaly detection accuracy. The use of blockchain further strengthens the system by ensuring data integrity and transparency. Overall, the Causio-TwinChain model offers a robust solution for building secure, trustworthy, and efficient digital twin environments.

2.6 Nguyen T., Kim D., Park J.: Blockchain in Digital Twin: A Comprehensive Survey

Context and Problem

The paper presents a detailed survey on the integration of blockchain technology with digital twin systems, focusing on how blockchain can enhance security, trust, and data management in digital environments. The context of this study lies in the increasing adoption of digital twins across domains such as smart manufacturing, healthcare, and smart cities, where large volumes of real-time data are generated and exchanged. The authors identify that traditional digital twin systems rely on centralized architectures, which are prone to data tampering, lack of transparency, and single points of failure. The key problem addressed is the absence of a secure, decentralized, and scalable mechanism to ensure data integrity and trust among multiple stakeholders interacting within digital twin ecosystems.

Methodology and Contribution

To address these issues, the paper adopts a systematic survey methodology, where existing research works and frameworks combining blockchain and digital twins are analyzed and compared. The authors categorize different integration architectures based on their design approaches, such as fully decentralized models, hybrid models combining cloud and blockchain, and edge-based implementations. The study evaluates these models using parameters like security, scalability, latency, and interoperability. The contribution of this paper lies in providing a structured classification of blockchain-enabled digital twin architectures, identifying key design

patterns, and highlighting open research challenges. It also outlines how blockchain features such as immutability, decentralization, and smart contracts can be effectively utilized to improve trust management in digital twin systems.

Key Findings

- Blockchain significantly enhances data integrity and prevents unauthorized modifications in digital twin systems.
- Decentralized architectures eliminate single points of failure and improve system reliability.
- Smart contracts enable automated trust management and secure data sharing, hybrid architectures offer a balance between performance and security.
- Scalability and latency remain critical challenges in large-scale implementations.

Conclusion

The paper concludes that the integration of blockchain with digital twin systems offers a promising solution to address trust and security challenges in modern digital infrastructures. However, it emphasizes the need for further research to overcome limitations related to scalability, computational overhead, and real-time processing. The study suggests that future developments should focus on optimizing consensus mechanisms, improving interoperability, and integrating advanced technologies such as artificial intelligence to further enhance trust evaluation and system efficiency.

2.7 Sharma P., Gupta R., Verma S.: Blockchain-Based Digital Twin for Industrial IoT Security

Context and Problem

The rapid adoption of Industrial Internet of Things (IIoT) technologies has enabled real-time monitoring and control of industrial systems through digital twins. However, these systems are highly vulnerable to cyber threats such as data tampering, unauthorized access, and denial-of-service attacks due to their reliance on interconnected devices and centralized architectures. The

primary problem addressed in this work is the lack of a secure and trustworthy mechanism to ensure the integrity and authenticity of data exchanged between physical devices and their corresponding digital twins in industrial environments.

Methodology and Contribution

To address these challenges, the authors propose a blockchain-integrated digital twin framework designed specifically for IIoT security. The methodology involves incorporating a decentralized blockchain network to record and validate all data transactions between sensors, devices, and digital twins. Each data entry is verified using consensus mechanisms before being added to the blockchain, ensuring immutability and preventing unauthorized modifications. Smart contracts are utilized to automate trust evaluation, enforce access control policies, and validate device behavior in real time. The system also includes identity management techniques to authenticate devices and prevent malicious nodes from participating in the network. The key contribution of this work lies in providing a secure, decentralized architecture that enhances trust, transparency, and resilience in industrial digital twin systems without significantly increasing computational overhead.

Key Findings

- Blockchain ensures tamper-proof storage of IIoT data and improves data integrity, smart contracts enable automated trust evaluation and access control enforcement.
- Decentralized architecture reduces dependency on centralized systems and eliminates single points of failure.
- Improved detection and prevention of malicious activities within the network.
- Enhanced reliability and security of communication between physical systems and digital twins.

Conclusion

The paper concludes that integrating blockchain technology with digital twin systems offers a robust solution for addressing security and trust issues in IIoT environments. The proposed framework successfully enhances system reliability, ensures secure data exchange, and supports

scalable industrial applications, making it a viable approach for next-generation smart manufacturing systems.

2.8 Li X., Zhang Y., Chen Z.: Trust Management in Blockchain-Based IoT Systems

Context and Problem

The rapid expansion of Internet of Things (IoT) environments has introduced significant challenges in managing trust among interconnected devices. In traditional IoT systems, trust is often maintained using centralized authorities, which creates vulnerabilities such as single points of failure, lack of transparency, and susceptibility to malicious attacks. With the integration of blockchain, IoT systems move toward decentralization; however, the problem of effectively evaluating and managing trust among distributed nodes still persists. The key issue addressed in this work is how to design a scalable and reliable trust management mechanism that can operate efficiently in a decentralized blockchain-based IoT environment while identifying malicious or unreliable nodes.

Methodology and Contribution

To address this problem, the authors propose a blockchain-based trust management framework that combines decentralized ledger technology with dynamic trust evaluation models. The methodology involves collecting behavioral data from IoT nodes, such as transaction history, communication patterns, and service reliability. These parameters are used to compute trust scores for each node using a reputation-based model. Blockchain is utilized to store trust values and transaction records in an immutable and transparent manner, ensuring that trust information cannot be tampered with. Smart contracts are employed to automate trust evaluation and update mechanisms, allowing the system to dynamically adjust trust scores based on node behavior. The contribution of this work lies in developing a scalable and decentralized trust management system that enhances reliability without relying on a central authority.

Key Findings

- Improved accuracy in identifying malicious and unreliable IoT nodes, enhanced transparency and immutability of trust records using blockchain.
- Dynamic trust evaluation adapts to changing node behaviour and reduction in false trust assignments compared to traditional models.

Conclusion

The paper concludes that integrating blockchain with trust management mechanisms provides a robust and transparent solution for decentralized IoT systems. By leveraging immutable ledgers and automated smart contracts, the proposed approach ensures reliable trust evaluation, improves system security, and supports the scalability required for large-scale IoT deployments.

2.9 Kumar V., Reddy P., Singh M.: AI-Based Trust Evaluation Models in Distributed Systems

Context and Problem

The rapid growth of distributed systems such as cloud computing, peer-to-peer networks, and IoT environments has introduced significant challenges in maintaining trust among participating entities. In such systems, nodes operate independently and interact dynamically, making it difficult to ensure the reliability and authenticity of data and services. Traditional trust models, which rely on static rules or predefined thresholds, are not effective in these environments because they cannot adapt to changing behaviors or detect sophisticated malicious activities. The key problem addressed in this work is the lack of an adaptive and intelligent trust evaluation mechanism capable of handling dynamic, large-scale, and heterogeneous distributed systems.

Methodology and Contribution

To address this issue, the authors propose an AI-based trust evaluation framework that utilizes both supervised and unsupervised machine learning techniques. The methodology involves collecting system behavior data such as communication patterns, transaction history, response time, and error rates from different nodes in the network. This data is then processed using machine

learning models to identify normal and abnormal patterns. Supervised learning algorithms are used to classify nodes as trustworthy or untrustworthy based on labeled historical data, while unsupervised learning techniques such as clustering are employed to detect anomalies without prior knowledge. A dynamic trust score is assigned to each node based on its behavior over time, and this score is continuously updated as new data is received. The main contribution of this work lies in introducing an adaptive trust model that can learn from system behavior, improve over time, and respond effectively to new threats.

Key Findings

- AI-based models significantly outperform traditional static trust models in dynamic environments.
- Real-time anomaly detection helps in identifying malicious nodes early, continuous trust score updating improves accuracy and adaptability.
- Combination of supervised and unsupervised learning enhances robustness, reduction in false positives compared to rule-based systems.

Conclusion

The study concludes that AI-based trust evaluation provides a scalable, flexible, and efficient solution for managing trust in distributed systems. By leveraging machine learning techniques, the system can dynamically adapt to changing conditions, detect malicious behavior more accurately, and ensure reliable interactions among nodes. This approach is particularly suitable for modern decentralized environments where traditional trust mechanisms are insufficient.

2.10 Patel N., Shah K., Mehta R.: Secure Digital Twin Framework Using AI and Blockchain

Context and Problem

The increasing adoption of digital twin technology across domains such as industrial automation, healthcare, and smart infrastructure has introduced significant concerns regarding data security, integrity, and trust. Digital twins rely on continuous data streams from physical systems, which

makes them vulnerable to data manipulation, cyber-attacks, and unauthorized access. Traditional centralized systems fail to provide adequate protection against these threats and lack transparency in data handling. The core problem addressed in this work is the absence of a unified and reliable framework that can ensure secure data exchange, real-time trust evaluation, and system integrity in digital twin environments.

Methodology and Contribution

To address these challenges, the authors propose a secure digital twin framework that integrates Artificial Intelligence (AI) and Blockchain technology. The methodology involves collecting real-time data from IoT devices and processing it through AI-based models for anomaly detection and trust evaluation. Machine learning algorithms analyze data patterns, identify inconsistencies, and assign trust scores to incoming data. Only data that meets predefined trust thresholds is considered valid. This validated data is then stored on a blockchain network, ensuring immutability, transparency, and decentralized control. Smart contracts are used to automate validation processes and enforce security policies. The contribution of this work lies in providing a comprehensive, real-time, and automated trust management system that enhances both security and operational efficiency in digital twin systems.

Key Findings

- Integration of AI improves accuracy in detecting anomalies and malicious data.
- Blockchain ensures tamper-proof data storage and decentralized trust, real-time trust evaluation enhances decision-making reliability.
- Smart contracts automate validation and reduce manual intervention, overall system security and data integrity are significantly improved.

Conclusion

The paper concludes that combining AI with blockchain creates a robust and scalable framework for secure digital twin systems. This integrated approach effectively addresses the limitations of traditional architectures by ensuring trusted data flow, enhancing system transparency, and enabling reliable decision-making. It also highlights the potential for further improvements

through advanced AI models and optimization of blockchain performance for large-scale deployments.

2.11 Yong Xiao, Dusit Niyato, Ping Wang, Dong In Kim, and Zhu Han: Toward Smart Cities: Digital Twin-Enabled Internet of Things With Blockchain and Artificial Intelligence

Context and Problem

With the rapid growth of smart cities, large-scale urban environments increasingly depend on interconnected Internet of Things (IoT) devices to monitor infrastructure, transportation, energy usage, and public services. These systems generate massive volumes of real-time data that are utilized by digital twins to simulate and optimize city operations. However, the authors identify several critical challenges in such environments. Traditional IoT-based smart city architectures are often centralized, making them vulnerable to data tampering, single points of failure, lack of transparency, and trust issues among multiple stakeholders**.** Additionally, ensuring the reliability of data collected from heterogeneous and potentially untrusted IoT devices becomes a major concern. The absence of an intelligent trust evaluation mechanism further complicates decision-making processes within digital twin systems.

Methodology and Contribution

The methodology involves creating digital twins of physical city components such as traffic systems, energy grids, and infrastructure, which continuously receive real-time data from IoT devices. Blockchain technology is incorporated to provide a decentralized and tamper-proof data storage mechanism, ensuring data integrity, traceability, and secure sharing across different entities. Artificial Intelligence is utilized to enhance the system's intelligence by performing data analysis, anomaly detection, and trust evaluation. AI models analyze incoming data streams to identify abnormal patterns, predict system behavior, and assign trust scores to data sources. The key contribution of this work lies in proposing a holistic architecture that seamlessly integrates these three technologies to improve security, trust, and efficiency in smart city environments.

Key Findings

- Integration of blockchain eliminates single points of failure and ensures data immutability, AI-based models significantly improve anomaly detection and predictive accuracy.
- Digital twins enable real-time monitoring and optimization of urban systems, the combined framework enhances trust among multiple stakeholders in smart city ecosystems.

Conclusion

The paper concludes that the integration of digital twin technology with blockchain and artificial intelligence provides a powerful solution for addressing trust, security, and efficiency challenges in smart cities. By leveraging blockchain for secure data management and AI for intelligent trust evaluation, the proposed system ensures reliable and transparent operations. The study highlights that such integrated frameworks are essential for the future development of scalable, secure, and intelligent smart city infrastructures, although further research is needed to address challenges related to computational complexity and large-scale deployment.

2.12 M. Grieves and J. Vickers: Digital Twin - Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems

Context and Problem

This paper is one of the foundational works that formally introduces and explains the concept of the digital twin. The authors focus on the problem of unpredictable and undesirable emergent behavior in complex systems such as aerospace systems, manufacturing environments, and large-scale engineering infrastructures. These systems consist of multiple interconnected components, and their interactions often produce behaviors that are difficult to predict using traditional modeling approaches. The lack of real-time visibility and continuous feedback between physical and virtual systems leads to inefficiencies, unexpected failures, and increased operational risks.

Methodology and Contribution

To address this problem, the authors propose the Digital Twin paradigm, which establishes a continuous and bidirectional connection between a physical system and its virtual counterpart. The methodology involves three core components: the physical entity, the digital model, and the data connection that links them. Real-time data collected from sensors is transmitted to the digital model, which simulates system behavior and provides insights into performance, potential failures, and optimization opportunities. The contribution of this work lies in defining a structured framework for digital twins and demonstrating how continuous data synchronization can reduce uncertainty in complex systems.

Key Findings

- Digital twins enable real-time monitoring and predictive analysis of complex systems, continuous data flow improves system visibility and reduces uncertainty.
- Early detection of anomalies helps prevent system failures and virtual simulation allows testing and optimization without affecting physical systems.

Conclusion

The paper concludes that the digital twin concept is a powerful approach to managing complexity in modern systems. By enabling real-time interaction between physical and virtual environments, it significantly reduces unpredictable behavior and improves system reliability, efficiency, and decision-making.

2.13 H. Ning, H. Liu, and J. Ma: Cyber-Physical-Social Based Security Architecture for Future Internet of Things

Context and Problem

This paper presents a forward-looking perspective on securing next-generation Internet of Things (IoT) systems by introducing a Cyber-Physical-Social (CPS) based security architecture**.** The authors recognize that traditional IoT security models mainly focus on cyber and physical

components, often neglecting the social dimension, which includes human behavior, social interactions, and user-generated data. The context of the study is the increasing complexity of IoT ecosystems, where devices, humans, and digital systems interact continuously. The key problem identified is that existing security frameworks are insufficient to handle multi-dimensional threats, particularly those arising from human-centric interactions such as social engineering, behavioral manipulation, and trust exploitation.

Methodology and Contribution

To address these challenges, the authors propose a CPS-based security architecture that integrates cyber space (digital systems), physical space (devices and sensors), and social space (human behavior and interactions). The methodology involves designing a layered security framework where each domain is protected individually while also ensuring coordinated security across domains. The architecture incorporates trust management, identity authentication, access control, and behavior analysis as core components. A significant contribution of this work is the introduction of social-aware security mechanisms, where user behavior and interaction patterns are analyzed to enhance trust evaluation and threat detection. The framework also emphasizes data fusion, combining information from cyber, physical, and social domains to improve decision-making and system resilience.

Key Findings

- Integration of cyber, physical, and social domains significantly improves overall security coverage.
- Social behavior analysis enhances trust evaluation and helps detect insider threats.
- Multi-layered architecture provides better resistance against complex and coordinated attacks.
- Data fusion across domains increases accuracy in threat detection and response.

Conclusion

The paper concludes that future IoT systems require a holistic security approach that goes beyond traditional models by incorporating social dimensions. The proposed CPS-based architecture

provides a comprehensive framework for addressing emerging security challenges and establishes a foundation for advanced trust evaluation mechanisms in complex interconnected systems.

2.14 K. Zhang, Y. Mao, S. Leng, Y. He, and Y. Zhang.: Mobile Edge Computing for Intelligent IoT Systems: Trust and Security Perspective

Context and Problem

This paper examines the role of mobile edge computing (MEC) in enabling intelligent IoT systems while addressing critical concerns related to trust and security. With the rapid growth of IoT devices, massive amounts of data are generated at the network edge, requiring efficient processing with minimal latency. However, traditional cloud-centric models introduce delays and create centralized points of vulnerability. The authors identify that moving computation to the edge improves performance but also introduces new challenges such **as** untrusted edge nodes, data privacy risks, and heterogeneous device environments. The core problem addressed is how to ensure secure, trustworthy, and efficient data processing in distributed edge-based IoT systems.

Methodology and Contribution

To address these challenges, the authors propose a framework that integrates mobile edge computing with trust management and security mechanisms. The methodology involves deploying computation and storage resources at the edge of the network, closer to IoT devices, thereby reducing latency and improving responsiveness. The framework incorporates trust evaluation models to assess the reliability of edge nodes and devices based on behavior, historical interactions, and data consistency. Security mechanisms such as encryption, authentication, and access control are integrated to protect data during transmission and processing. The contribution of this work lies in providing a comprehensive architecture that combines low-latency edge computing with robust trust and security strategies, making it suitable for intelligent IoT applications.

Key Findings

- Mobile edge computing significantly reduces latency and improves real-time data processing.
- Trust evaluation mechanisms enhance reliability of edge nodes and IoT devices.

- Integration of security techniques mitigates risks such as data tampering and unauthorized access.

Conclusion

The paper concludes that mobile edge computing, when combined with effective trust and security mechanisms, provides a scalable and efficient solution for intelligent IoT systems. It highlights the importance of incorporating adaptive trust evaluation models to handle dynamic and heterogeneous environments, which is highly relevant for advanced systems such as blockchain-based digital twins.

2.15 Q. Xu, Z. Su, and Y. Wang: Blockchain-Based Trust Management in Distributed Systems: A Survey

Context and Problem

With the rapid growth of distributed systems such as IoT networks, cloud computing platforms, and decentralized applications, establishing trust among participating entities has become a fundamental challenge. In highly dynamic environments, these static models fail to adapt to changing behaviors of nodes, making them ineffective in detecting malicious activities. The paper identifies that while blockchain technology offers decentralization, immutability, and transparency, integrating it effectively with trust management systems is still an open research problem. Specifically, challenges such as scalability, computational overhead, and efficient trust evaluation mechanisms need to be addressed. The authors aim to systematically analyze existing blockchain-based trust management approaches and identify gaps in current research.

Methodology and Contribution

The authors adopt a comprehensive survey-based methodology, where they review and categorize existing trust management models that incorporate blockchain technology. The study examines different architectural designs, consensus mechanisms, and trust evaluation techniques used in distributed systems. It classifies trust models based on parameters such as reputation systems, behavior-based evaluation, and hybrid approaches combining multiple trust factors. A significant contribution of this paper is the taxonomy of blockchain-based trust management systems, which

helps in understanding how trust is established, maintained, and updated in decentralized environments. The authors also analyze the role of smart contracts in automating trust evaluation and enforcing security policies.

Key Findings

- Blockchain eliminates dependency on centralized trust authorities by enabling decentralized validation.
- Reputation-based and behavior-based trust models are widely used but have scalability limitations.
- High computational and storage overhead remains a major limitation in blockchain-based systems.

Conclusion

The paper concludes that blockchain technology provides a strong foundation for building secure and transparent trust management systems in distributed environments. However, existing solutions still face challenges related to scalability, efficiency, and real-time processing. The authors emphasize the need for hybrid approaches that combine blockchain with AI and lightweight consensus mechanisms to improve performance. This survey serves as a valuable reference for designing advanced trust evaluation systems, particularly in applications such as digital twins, where reliable and secure data exchange is critical.

2.16 Y. Lu: Digital Twin-Driven Smart Manufacturing: Connotation, Reference Model, Applications and Research Issues

Context and Problem

This paper presents a comprehensive understanding of digital twin technology in the context of smart manufacturing. The author emphasizes that modern manufacturing systems require high levels of flexibility, real-time monitoring, and intelligent decision-making. However, traditional manufacturing systems face limitations such as lack of real-time synchronization, inefficient data utilization, and absence of a unified framework for integrating physical and virtual systems. The

key problem identified is the absence of a standardized architecture and clear conceptual understanding of digital twin-driven manufacturing, which restricts its large-scale adoption.

Methodology and Contribution

To address these issues, the author proposes a reference model for digital twin-driven smart manufacturing. The methodology is based on defining the core components of a digital twin system, including the physical entity, virtual model, data processing layer, and service system. The paper introduces a structured framework that enables seamless interaction between physical and digital spaces through continuous data exchange. It also highlights the role of enabling technologies such as IoT, big data analytics, cloud computing, and artificial intelligence in building intelligent manufacturing systems. The major contribution of this work lies in formalizing the concept of digital twin in manufacturing and providing a standardized reference architecture that can guide future implementations.

Key Findings

- Establishes a clear definition and structure for digital twin-driven manufacturing, identifies key enabling technologies such as IoT, AI, and cloud computing.
- Demonstrates improved efficiency through real-time monitoring and predictive analysis.
- Highlights the importance of data integration between physical and virtual systems.

Conclusion

The paper concludes that digital twin technology has the potential to transform traditional manufacturing into intelligent, data-driven systems. However, challenges such as system integration, data management, and standardization need to be addressed to fully realize its benefits. The proposed reference model serves as a foundational framework for future research and development in smart manufacturing systems.

Table – 2.1: Summary of Literature Survey on AI Assisted Trust Evaluation In Blockchain Based Digital Twin System

No	Title	Authors	Problem Addressed	Methodology	Key Findings	Revelance
1.	Blockchain-Based Trust Mechanism for Digital Twin in IIoT	Zhang Y., Liu X., Wang H.	Lack of trust and data integrity in IIoT-based digital twins	PoA-based blockchain integrated with trust evaluation parameters	Improved data integrity, reduced overhead, enhanced trust	Core foundation for blockchain-based trust in digital twins
2.	TFL-DT: Trust Evaluation Scheme for Federated Learning in Digital Twin	Chen J., Li Q., Zhao K.	Malicious participants in federated learning affecting trust	AI-based multi-factor trust scoring with dynamic updates	Improved model accuracy, detection of malicious nodes	Shows AI-based trust evaluation mechanism
3.	Blockchain-Enabled Digital Twin for Smart Cities	Kumar A., Singh R., Patel S.	Data inconsistency and lack of transparency in smart cities	LSTM-based prediction + blockchain storage	Better prediction, increased transparency and trust	Demonstrates AI + blockchain integration in real-world systems
4.	Blockchain-Based Collaborative Digital Twin Framework for Healthcare	Rahman M., Islam S., Ahmed T.	Secure data sharing in healthcare digital twins	Blockchain with encryption and access control	Secure collaboration, reduced data breaches	Highlights trust in sensitive environments
5.	Causio-TwinChain: AI-Driven Trust Model	Wang L., Zhou Y., Chen H.	High false positives in anomaly detection	Causal AI + blockchain for trust evaluation	Reduced false positives, improved accuracy	Advanced AI-based trust evaluation approach
6.	Blockchain in Digital Twin: A Comprehensive Survey	Nguyen T., Kim D., Park J.	Lack of unified understanding of DT + blockchain systems	Comparative analysis of architectures	Identified scalability and efficiency issues	Provides theoretical background and research gaps
7.	Blockchain-Based Digital Twin for Industrial IoT Security	Sharma P., Gupta R., Verma S.	Cyber-attacks and data tampering in industrial systems	Blockchain + smart contracts for validation	Improved security, automated trust enforcement	Industrial application of secure digital twins

8.	Trust Management in Blockchain-Based IoT Systems	Li X., Zhang Y., Chen Z.	Inefficient traditional trust models in decentralized systems	Trust model using behavior, history, and reputation	Better trust accuracy and transparency	Fundamental trust model applicable to DT systems
9.	AI-Based Trust Evaluation Models in Distributed Systems	Kumar V., Reddy P., Singh M.	Static trust models not suitable for dynamic systems	ML-based anomaly detection and adaptive trust scoring	Improved adaptability and detection efficiency	Supports AI-based trust evaluation concept
10.	Secure Digital Twin Framework Using AI and Blockchain	Patel N., Shah K., Mehta R.	Lack of unified secure framework for DT systems	AI anomaly detection + blockchain storage + smart contracts	Real-time trust evaluation, improved security	Directly aligns with proposed seminar topic
11.	Digital Twin-enabled IoT with Blockchain and AI	Wan et al.	Lack of secure and intelligent IoT systems for smart cities	Integration of AI, blockchain, and digital twin for smart city framework	Improved system intelligence, security, and real-time decision-making	High system complexity and scalability issues
12.	Digital Twin for Complex Systems	Grieves & Vickers	Unpredictable behavior in complex physical systems	Conceptual digital twin model for monitoring and prediction	Enhanced system understanding and predictive capability	Lacks implementation details and security mechanisms
13.	Security Architecture for IoT Systems	Ning et al.	Security vulnerabilities in IoT environments	Cyber-physical-social security architecture	Improved system security and trust management	Does not include blockchain integration
14.	Trust and Security in Edge Computing for IoT	Zhang et al.	Trust issues in distributed IoT and edge environments	Edge computing with trust evaluation mechanisms	Reduced latency and improved trust handling	Limited focus on blockchain-based security
15.	Blockchain-based Trust Management Survey	Xu et al.	Inefficient trust management in distributed systems	Survey of blockchain-based trust mechanisms	Highlights effectiveness of decentralized trust models	Mostly theoretical, lacks practical implementation
16.	Digital Twin-driven Smart Manufacturing	Lu Y.	Inefficiency in traditional manufacturing systems	Digital twin-based smart manufacturing model	Improved efficiency, monitoring, and automation	Does not address trust and security in depth

CHAPTER 3

PROBLEM/TOPIC ANALYSIS AND DESIGN

3.1 Problem Definition

Digital Twin systems are increasingly used to replicate physical systems in a virtual environment for monitoring, simulation, and predictive analysis. These systems rely on continuous streams of data collected from sensors and IoT devices. However, the correctness and reliability of the digital twin depend entirely on the quality and trustworthiness of the input data. If compromised or inaccurate data is introduced into the system, it can lead to incorrect analysis and potentially harmful decisions.

One of the major challenges in such environments is the lack of dynamic trust evaluation mechanisms. Traditional systems assume that incoming data is reliable, which is not always valid in distributed and open environments. Furthermore, centralized architectures used in many digital twin implementations are vulnerable to attacks, data tampering, and system failures. This creates a critical need to analyze how trust can be established, evaluated, and maintained in such systems.

3.2 Analysis of Trust Issues in Digital Twin Systems

In digital twin environments, trust issues arise due to multiple factors such as heterogeneous data sources, unreliable devices, and malicious entities. Since data is collected from various sensors and external systems, there is always a possibility of data inconsistency, spoofing, or manipulation. Additionally, insider threats and compromised devices can inject false data into the system.

Another key issue is the absence of transparency in traditional systems. Users cannot verify whether the data being used has been altered or whether it originates from a trusted source. This lack of traceability reduces confidence in the system. Moreover, static trust models fail to adapt to dynamic environments where device behavior and data patterns change over time.

3.3 Need for Blockchain in Digital Twin Systems

Blockchain technology provides a decentralized approach to address many of the limitations observed in traditional digital twin systems. By storing data in a distributed ledger, blockchain ensures that once data is recorded, it cannot be modified or deleted. This property of immutability enhances data integrity and prevents tampering.Additionally, blockchain improves transparency and traceability, allowing all participants in the network to verify the authenticity of data.

The following figure 3.3.1 illustrates the conceptual architecture of a digital twin system, showing the interaction between physical processes, communication interfaces, data aggregation, and analysis components

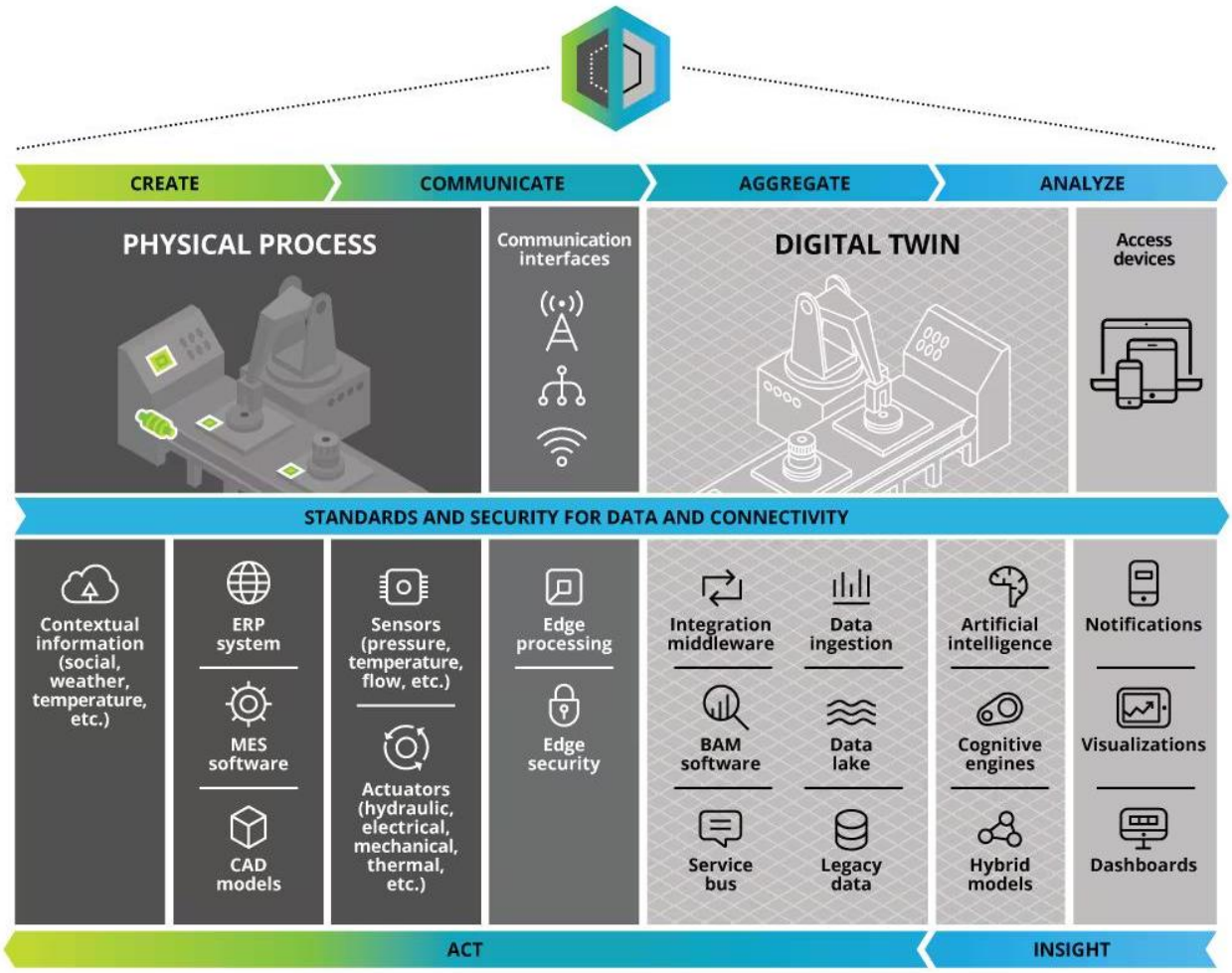

Fig 3.3.1 Conceptual Architecture of Digital Twin System

3.4 Role of AI in Trust Evaluation

Artificial Intelligence introduces the capability to analyze large volumes of data and identify patterns that may indicate abnormal or malicious behavior. In the context of digital twin systems, AI can be used to evaluate the trustworthiness of incoming data by detecting anomalies, inconsistencies, and deviations from expected behavior.

Machine learning models can be trained on historical data to recognize normal patterns and identify suspicious activities. Based on this analysis, the system can assign **trust scores** to data sources or individual data points. This dynamic approach allows the system to adapt to changing conditions and continuously improve its trust evaluation process. However, AI alone cannot guarantee data security or prevent tampering after evaluation.

The following figure 3.4.1 illustrates the detection of anomalies in data using an AI-based approach, where abnormal patterns are identified and assigned anomaly scores. This demonstrates how trust evaluation can be performed by distinguishing normal behavior from suspicious or inconsistent data.

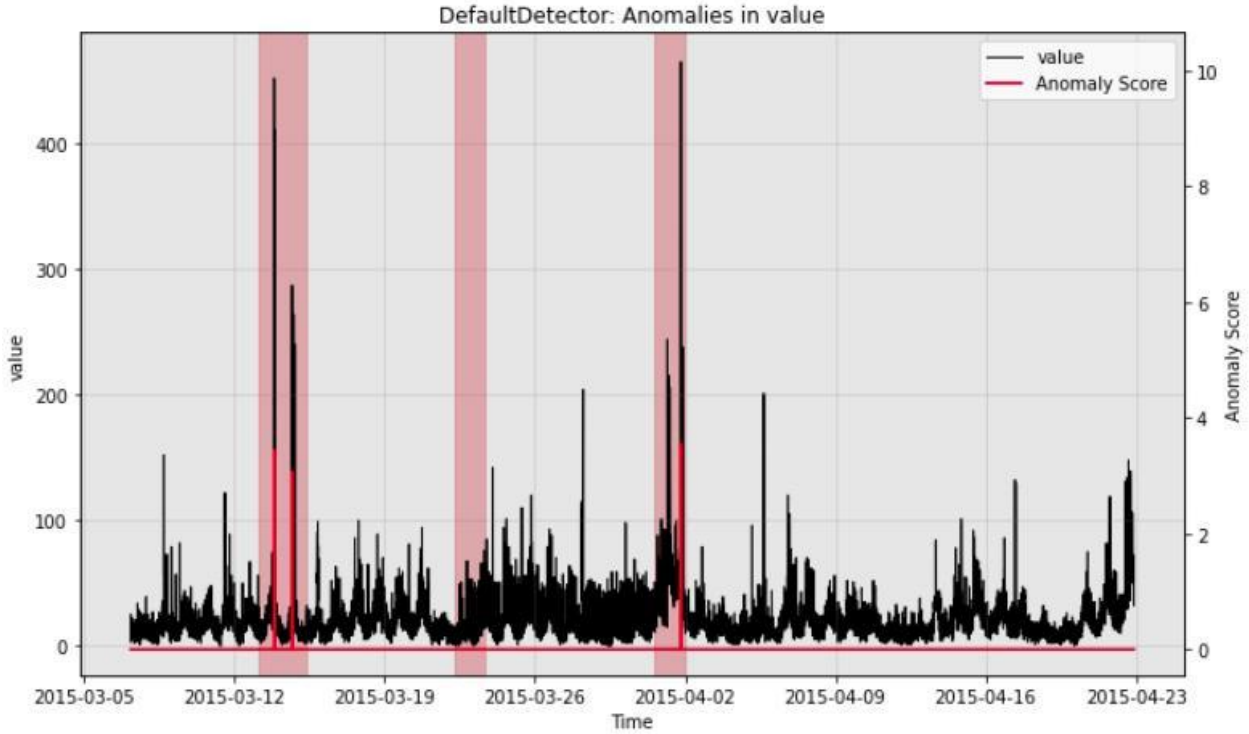

Fig 3.4.1 Anomaly Detection Using AI- Based Trust Evaluation

3.5 Combined Analysis: AI and Blockchain Integration

The integration of AI and blockchain addresses the limitations of using each technology independently. AI provides intelligent trust evaluation by analyzing data before it is accepted into the system, while blockchain ensures that once data is validated, it is stored securely and cannot be altered. In this combined approach, data collected from sensors is first processed by AI models to assess its reliability. Only data that meets certain trust criteria is recorded on the blockchain. This ensures that the digital twin operates on verified and trustworthy data. The integration also enables automated validation through smart contracts, improving efficiency and reducing manual intervention.

The following figure 3.5.1 illustrates the integration of edge, blockchain, and cloud layers in a digital twin system for secure data processing and storage.

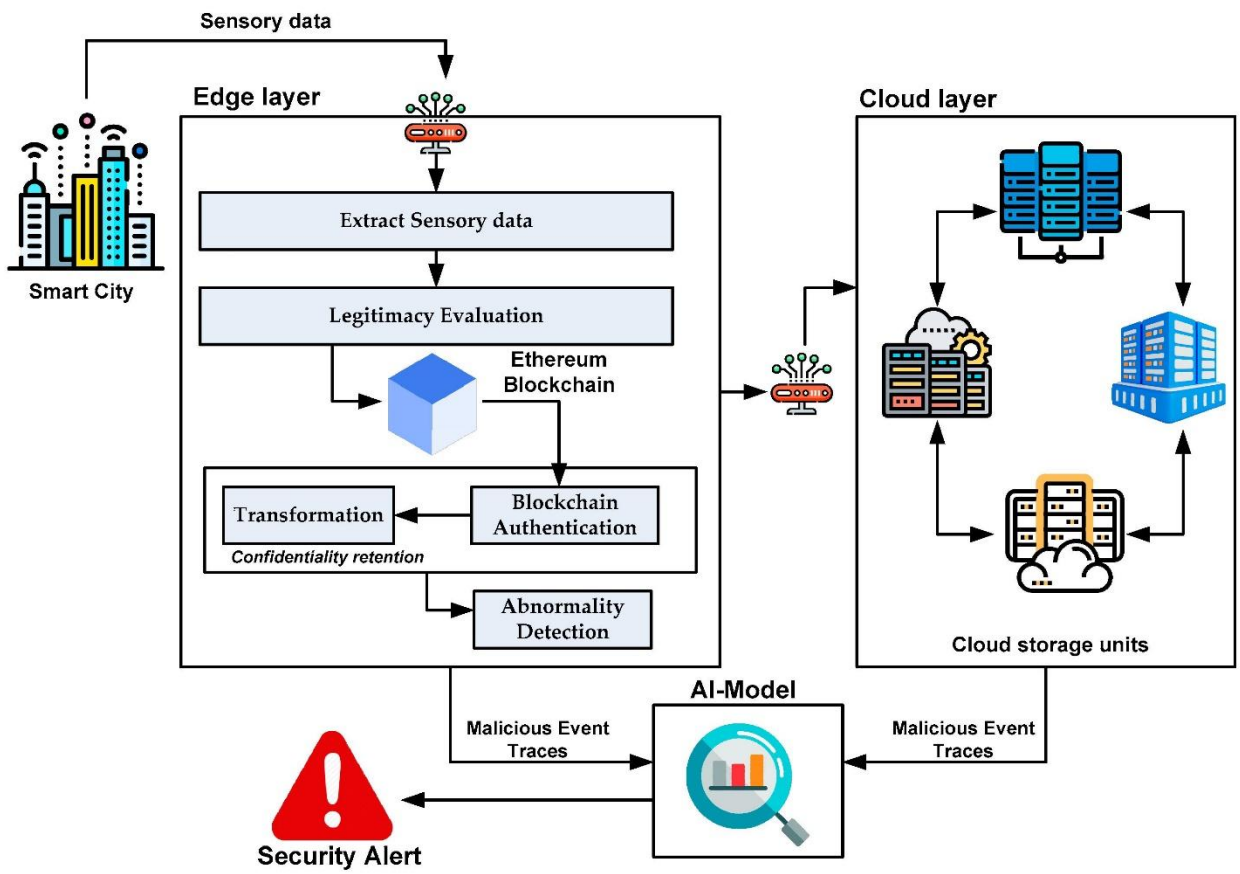

Fig 3.5.1 AI and Blockchain Integrated Digital Twin Architecture

3.6 System Design

The following figure 3.6.1 represents how different technologies AI, Blockchain, and Digital Twin are logically organized and interact with each other to ensure trusted and secure system operation.

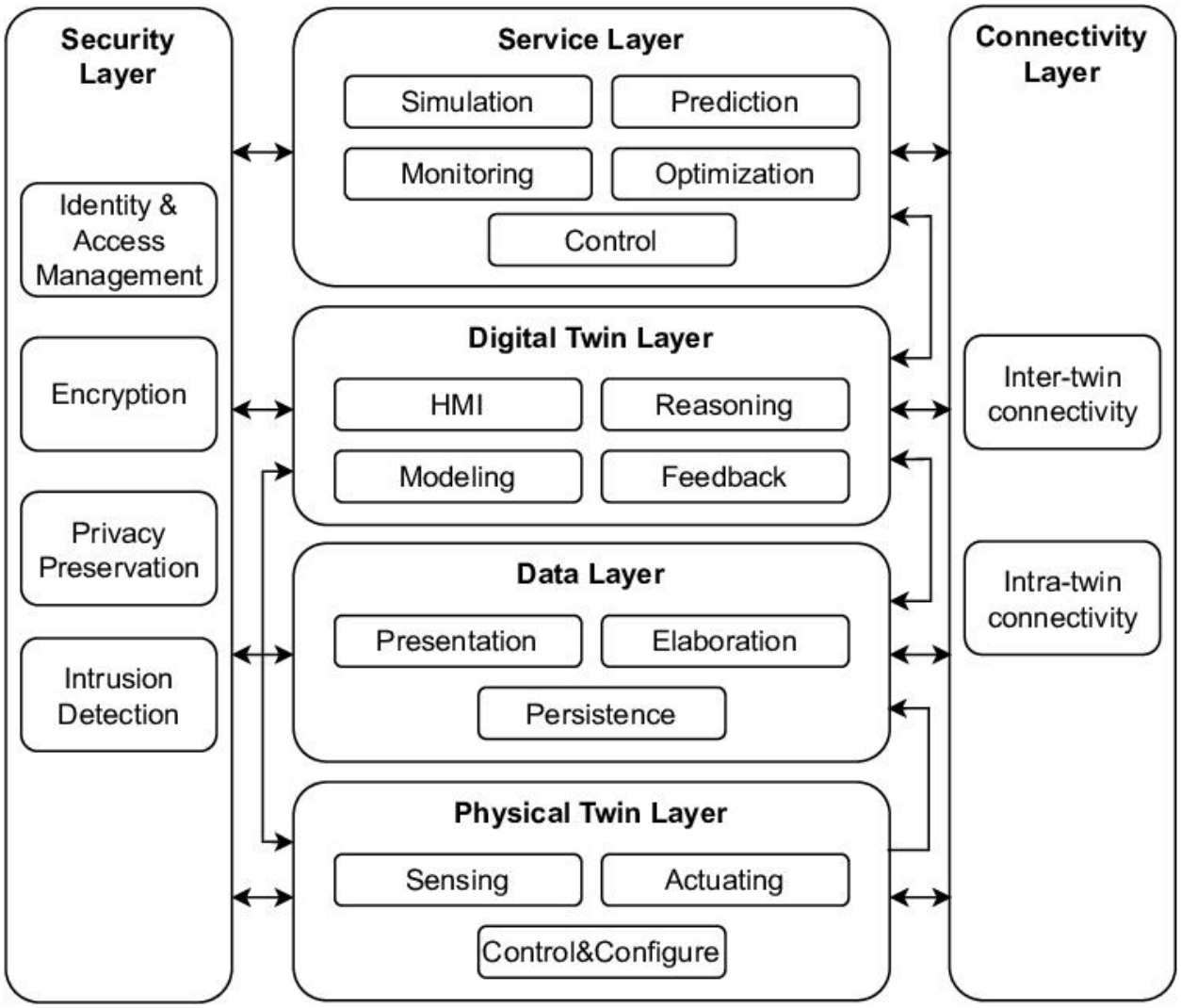

Fig 3.6.1 Layered Architecture of Digital Twin System

At the lowest level of the system, the physical layer consists of real-world entities such as machines, sensors, medical devices, or infrastructure components. These devices continuously generate data such as temperature, pressure, location, or operational status. This data forms the foundation of the entire system, but at this stage, it is raw and unverified, meaning it cannot yet be trusted for critical decisions.

The next level is the data acquisition and communication layer, where data is transmitted from sensors to processing units using communication technologies such as IoT protocols, wireless networks, or cloud gateways. During this transmission, data may be exposed to risks such as interception, manipulation, or loss. Therefore, this layer must ensure reliable and efficient data transfer.

Above this lies the AI-based trust evaluation layer, which plays a crucial role in the system. Here, machine learning models analyze incoming data to determine whether it is trustworthy. The system checks for anomalies, inconsistencies, or unusual patterns by comparing incoming data with historical behavior. For example, if a sensor suddenly reports values far outside its normal range, the AI model may classify it as suspicious and assign a low trust score. Only data that meets predefined trust criteria is allowed to move forward in the system.

Once the data is validated, it enters the blockchain layer, where it is securely stored in a decentralized ledger. Blockchain ensures that the data is immutable, meaning it cannot be altered once recorded. It also provides transparency and traceability, allowing all participants in the network to verify the authenticity of the data. Smart contracts within the blockchain automate validation rules and enforce security policies without human intervention.

At the top of the architecture is the digital twin layer, which uses the verified and trusted data to create a virtual representation of the physical system. This layer performs simulations, predictive analysis, and optimization. Since the data used here has already passed through trust evaluation and secure storage, the digital twin can operate with high accuracy and reliability.

3.7 Design Considerations

- **Scalability:** Digital twin systems may involve thousands or even millions of devices generating continuous data streams. The system must be capable of handling this large volume of data without performance degradation. Therefore, selecting efficient consensus mechanisms or hybrid architectures becomes essential.
- **Latency and Real-time performance:** If trust evaluation or blockchain validation introduces delays, the system may fail to meet real-time requirements. Hence, the design

must balance speed and security, possibly by using lightweight AI models and faster blockchain frameworks.

- **Security:** This includes encryption, secure communication protocols, and protection against attacks such as spoofing or data injection. At the same time, privacy must be maintained, especially when dealing with sensitive data like patient information. This creates a trade-off between transparency (blockchain) and confidentiality.
- **Interoperability:** These components may use different standards and protocols, so the system must ensure seamless communication between them. Without proper interoperability, integration becomes complex and inefficient.
- **Accuracy and Reliability:** Poorly trained models may either reject valid data or accept malicious data, both of which can harm the system. Therefore, proper training, validation, and continuous updating of AI models are necessary.
- **Efficiency and Resource Utilization:** AI algorithms and blockchain operations can be computationally intensive, requiring significant processing power and energy. This becomes a challenge in resource-constrained environments such as edge devices. Optimizing algorithms and distributing workloads efficiently is essential to maintain system performance.
- **System Adaptability and Flexibility:** The system should be capable of adapting to new devices, changing data patterns, and evolving threats. A rigid design may quickly become outdated, whereas a flexible system can evolve with technological advancements.

3.8 Challenges Identified in the Design

The integration of AI and blockchain introduces several challenges. Blockchain networks can experience latency and scalability issues, particularly in large-scale deployments. AI models require significant computational resources and high-quality training data to function effectively.

Another challenge is maintaining a balance between transparency and privacy, as blockchain systems are inherently transparent while some applications require confidentiality. Ensuring seamless integration between heterogeneous components and maintaining system performance under high loads are also critical challenges.

CHAPTER 4

OUTCOMES & UNDERSTANDING

4.1 Understanding of Integrated Technologies

Digital twins provide a virtual representation of physical systems, blockchain ensures secure and tamper-proof data storage, and artificial intelligence enables intelligent analysis and trust evaluation. The integration of these technologies demonstrates how modern systems are no longer dependent on a single approach but rely on a combination of complementary solutions to achieve reliability, security, and efficiency. Digital twins act as the core operational layer, continuously reflecting the state of physical systems in a virtual environment. However, their effectiveness is entirely dependent on the accuracy and reliability of incoming data. This is where artificial intelligence becomes essential. AI does not just process data - it interprets patterns, detects anomalies, and evaluates the credibility of data sources, enabling the system to distinguish between normal and suspicious behavior.

Furthermore, the integration highlights the shift towards intelligent decentralized systems, where decisions are made based on verified data and automated processes. Smart contracts in blockchain can enforce rules, while AI continuously adapts to new data patterns, making the system both autonomous and scalable. Overall, this integrated approach demonstrates how combining digital twins, AI, and blockchain leads to systems that are not only secure and efficient but also adaptive, transparent, and capable of handling real-world uncertainties, which is essential for next-generation smart applications.

4.2 Improved Knowledge of Trust Evaluation Mechanisms

AI-based trust evaluation introduces a more adaptive and intelligent approach. Instead of relying on static rules, machine learning models continuously analyze incoming data and compare it with historical patterns. For example, if a sensor that typically reports stable temperature values suddenly produces extreme fluctuations, the AI model can recognize this as an anomaly and flag it as suspicious. Another key aspect is the concept of trust scoring, where each data input or data source is assigned a numerical value representing its reliability. This enables the system to make

more informed decisions, such as whether to accept, reject, or further analyze the data before using it.

Overall, this understanding demonstrates that trust evaluation in modern systems is no longer a one-time verification process but a continuous and evolving mechanism. AI enables systems to learn from past data, adapt to changing conditions, and improve their decision-making over time. This leads to more reliable, secure, and intelligent digital twin systems where decisions are based on validated and trustworthy data rather than assumptions.

4.3 Understanding the Role of Blockchain in Security

Blockchain technology plays a crucial role in enhancing the security of digital twin systems by providing a decentralized and tamper-resistant data management mechanism. Unlike traditional centralized systems, where data is stored in a single location and can be modified or attacked, blockchain distributes data across multiple nodes in a network. One of the key features of blockchain is immutability, which ensures that once data is recorded in the ledger, it cannot be altered or deletedAdditionally, blockchain provides transparency and traceability, allowing all participants to verify the origin and history of data, which is essential for building trust in distributed environments.

Furthermore, blockchain enhances security by incorporating cryptographic techniques, such as hashing and digital signatures, to protect data during storage and transmission. These mechanisms ensure data authenticity and prevent unauthorized access. Overall, the integration of blockchain in digital twin systems provides a robust security framework that ensures data integrity, transparency, and reliable trust management.

4.4 Insight into Real-Time Data Processing and Decision Making

Data generated from physical entities such as sensors and IoT devices is transmitted instantly to the system, where it undergoes preprocessing and analysis. This continuous flow of data enables the system to maintain an up-to-date representation of the physical environment within the digital twin. Artificial Intelligence plays a crucial role in this process by analyzing incoming data streams in real time to detect anomalies, evaluate trust, and filter unreliable inputs. Only data that meets

the required trust criteria is forwarded for further use. This ensures that decisions are not based on corrupted or suspicious data, thereby improving the overall reliability of the system.

Once the data is validated, it is securely stored using blockchain, ensuring integrity and traceability. The digital twin then utilizes this trusted data to perform simulations, predictive analysis, and optimization. This enables timely and accurate decision-making, which is essential in critical applications such as healthcare monitoring, industrial automation, and smart city management. Overall, the integration of real-time data processing with AI and blockchain enhances the system's ability to respond quickly to changes, improve operational efficiency, and support intelligent, data-driven decision-making.

4.5 Practical Understanding of Security and Privacy Trade-offs

Transparency can conflict with privacy requirements, especially in applications where sensitive data is involved, such as healthcare records or industrial data. In such systems, while it is important to ensure that data is trustworthy and cannot be tampered with, it is equally important to prevent unauthorized access to confidential information. This creates a challenge where increasing transparency may lead to reduced privacy, and enhancing privacy may limit visibility and traceability. To address this, techniques such as data encryption, access control mechanisms, and permissioned blockchains are used to restrict data visibility while still maintaining security.

Additionally, AI-based trust evaluation introduces another layer where sensitive data is analyzed, raising concerns about data exposure during processing. Therefore, it becomes essential to design systems that carefully balance these aspects by ensuring that only necessary information is shared, while sensitive data remains protected. This understanding highlights the importance of designing systems that can achieve both secure data handling and privacy preservation without compromising overall system performance.

4.6 Application-Oriented Understanding

The study provides a clear understanding of how AI-assisted trust evaluation in blockchain-based digital twin systems can be applied across various real-world domains where data reliability and security are critical. In smart cities, such systems can be used for traffic monitoring, energy management, and infrastructure maintenance by ensuring that only verified data is used for

decision-making. In the healthcare sector, digital twins can represent patient conditions, where trusted data from medical devices enables accurate diagnosis and continuous monitoring while maintaining data security and privacy.

In industrial IoT environments, this approach supports predictive maintenance by analyzing trusted sensor data to detect faults and prevent system failures. Similarly, in supply chain management, blockchain ensures transparency and traceability of goods, while AI evaluates the authenticity of transaction data, reducing fraud and improving trust among stakeholders. These applications demonstrate that integrating AI, blockchain, and digital twin technologies not only enhances system reliability but also enables efficient, secure, and intelligent operations across multiple domains.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

The study of AI-assisted trust evaluation in blockchain-based digital twin systems highlights the importance of ensuring data reliability, security, and intelligent decision-making in modern digital environments. By analyzing the integration of digital twins, blockchain, and artificial intelligence, it becomes evident that each technology addresses a specific challenge—digital twins enable real-time simulation, blockchain ensures data integrity, and AI provides dynamic trust evaluation. Together, they form a robust framework capable of supporting secure and efficient data-driven systems.

5.1 Summary of Key Insights

This work emphasizes that trust is a critical factor in digital twin systems, as their effectiveness depends on the accuracy and reliability of input data. The integration of AI enables intelligent detection of anomalies and dynamic trust scoring, allowing the system to filter out unreliable or malicious data before it is used. Blockchain complements this by ensuring that validated data is securely stored in an immutable and transparent manner, preventing tampering and enhancing accountability. Additionally, the study highlights how the combination of these technologies supports real-time decision-making, improves system resilience, and reduces dependence on centralized control. It also brings forward the importance of a layered architecture, where each component data collection, trust evaluation, secure storage, and simulation plays a distinct role in maintaining system integrity. Overall, the integration of AI, blockchain, and digital twins provides a comprehensive approach to building secure, scalable, and trustworthy digital systems in modern applications.

5.2 Significance of the Study

The study demonstrates the potential of combining emerging technologies to address real-world challenges in distributed systems. It provides a deeper understanding of how secure and trustworthy systems can be designed for applications such as smart cities, healthcare, and industrial

automation. This highlights the growing importance of interdisciplinary approaches in solving complex technological problems.

In addition, the study emphasizes the shift from traditional centralized systems to decentralized and intelligent architectures, where trust is not assumed but continuously evaluated. It also brings attention to the importance of data integrity in decision-critical environments, where even minor inaccuracies can lead to significant consequences. By exploring this integrated approach, the study contributes to the development of more resilient, transparent, and adaptive systems, which are essential for future digital transformation and Industry 4.0 applications.

5.3 Limitations

One of the primary concerns is the high computational complexity, as both AI models and blockchain operations require significant processing power and energy consumption. This can make the system less suitable for resource-constrained environments such as edge devices.

Another important limitation is latency, as the processes of trust evaluation and blockchain validation may introduce delays, which can affect real-time applications such as healthcare monitoring or industrial control systems. There is also a challenge in maintaining a balance between transparency and privacy, since blockchain systems are inherently transparent while certain applications require strict confidentiality of sensitive data.

Interoperability issues may arise due to the integration of multiple technologies with different standards and protocols. The complexity of system design and implementation also increases, requiring specialized expertise and infrastructure.

5.4 Future Scope

Future developments in AI-assisted trust evaluation for blockchain-based digital twin systems can focus on improving the scalability, efficiency, and adaptability of the overall framework. One important direction is the use of advanced AI techniques such as deep learning and federated learning, which can enhance trust evaluation by learning from distributed data sources without compromising privacy.

Further research can explore lightweight and scalable blockchain solutions, such as Layer-2 protocols or hybrid blockchain architectures, to overcome performance limitations. Enhancing interoperability between different platforms and standards will also be essential for large-scale adoption. Moreover, future systems can focus on self-adaptive and autonomous trust models that continuously learn and evolve based on changing system behavior and emerging threats. The integration of explainable AI (XAI) can also improve transparency in decision-making by making trust evaluation processes more understandable to users. Overall, continued advancements in these areas will contribute to building more secure, intelligent, and scalable digital twin ecosystems.

In conclusion, AI-assisted trust evaluation in blockchain-based digital twin systems represents a promising direction for building secure, intelligent, and reliable digital ecosystems. With continuous advancements and research, this approach has the potential to play a crucial role in the future of smart and connected systems.

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